Smart Toll Tax System

Sensor Lab mini Project

Third Year (Semester-VI) Bachelors in Information Technology, Autonomous Program

By

Student Name: Ria Chaudhari Class_Roll no: D15A_07
Student Name: Eesha Chavan Class_Roll no: D15A_08
Student Name: Shraeyaa Dhaigude Class_Roll no: D15A_15

Department of Information Technology
Vivekanand Education Society's Institute of Technology
An Autonomous Institute Affiliated to University of Mumbai

2023-2024

Certificate

This is to certify that Ria Chaudhari, Eesha Chavan, Shraeyaa Dhaigude have completed the project report on Dry and Wet Object Segregator satisfactorily in partial fulfilment of the requirements for the award of Mini Project in sensor lab of third Year, (Semester-VI) in Information Technology under the guidance of Mr. Abhishek Chaudhari during the year 2024-2025 as prescribed by An Autonomous Institute Affiliated to University of Mumbai

Supervisor/ Examiner

Table of Contents:

List of Figures:

Abstract

Traditional toll collection methods often result in long queues, time consumption, and human error, leading to inefficiencies in traffic flow and revenue collection. To address these issues, this project presents a **Smart Toll Tax System** utilizing **RFID technology and Arduino microcontrollers**. The system enables automatic identification of vehicles through RFID tags, allowing seamless toll collection without manual intervention. As a vehicle approaches the toll booth, the RFID reader detects the tag, verifies user credentials, and either permits access or prompts for recharge, all in real-time. Designed with affordability, accuracy, and scalability in mind, this solution is ideal for deployment on highways and in smart city infrastructures. The prototype demonstrates the potential for reducing congestion, increasing transparency, and modernizing the toll collection process with minimal human effort.

Chapter 1: Introduction

1.1 Background

With growing urban traffic, traditional toll collection methods face issues like delays, errors, and fuel wastage. Manual systems cause long queues, emissions, and driver frustration. While automated toll systems exist, they are often costly and complex—unsuitable for smaller towns or educational setups. This project proposes a low-cost, Arduino-based RFID toll collection system that offers real-time, efficient toll management. It is affordable, easy to implement, and ideal for learning or deployment in resource-limited areas.

1.2 Objectives

- **Automate Toll Collection Using RFID:** Design and implement a system that automatically identifies vehicles using RFID tags and deducts toll amounts without the need for manual intervention.

- **Enhance Efficiency and Reduce Congestion:** Improve the speed and accuracy of toll processing to minimize traffic buildup at toll booths, reduce human errors, and enhance the overall user experience.

- **Develop a Low-Cost, Scalable Solution:** Build a compact and economically viable prototype using Arduino and RFID technology that can be easily implemented in small-scale toll plazas or as a model in academic environments.

Chapter 2: Literature Review

2.1 Survey

1. RFID-Based Automated Toll Tax Collection Using Arduino:

In the paper *"RFID-Based Automated Toll Tax Collection Using Arduino"* (JETIR, 2020), researchers implemented an automated toll system that utilizes RFID technology integrated with Arduino. The system reduces human involvement by using RFID tags on vehicles for automatic toll deduction. It also helps minimize time delays and ensures smoother traffic flow at toll booths. The paper emphasizes cost-effectiveness and practical feasibility for small-scale deployments.

2. Automatic Toll Tax System Using Arduino:

The paper *"Automatic Toll Tax System Using Arduino"* (IJARSCT, 2021) proposes a simplified toll management solution using RFID cards and Arduino. It focuses on decreasing fuel wastage and vehicle congestion by enabling automatic gate access. The system detects vehicles using RFID and initiates gate operations through servo motors. It is suitable for rural or semi-urban toll booths with minimal infrastructure requirements.

3. Automated Toll Management System Using RFID and Image Processing:

In *"Automated Toll Management System Using RFID and Image Processing"* (arXiv, 2024), the authors combine RFID with image recognition techniques using CNN-based number plate detection. The hybrid model enhances system accuracy and security by double-checking vehicle identity. The paper demonstrates how this method can reduce fraud and make toll booths more intelligent through IoT integration.

4. Automatic Car Parking Toll Gate System Using Arduino:

The study titled *"Automatic Car Parking Toll Gate System Using Arduino"* (IJPREMS, 2025) explores automated toll collection for parking lots. It uses RFID to detect vehicle entry and exit while managing gate operations via Arduino-controlled servo motors. The paper outlines an efficient low-power design that is easy to implement and helps reduce manual monitoring in controlled environments like campuses or apartment complexes.

2.2 Research Gap

Despite advancements in automated waste segregation, several challenges remain unaddressed:

- **Application-Specific Limitations:** Existing systems are often tailored to specific environments, such as agricultural fields or urban households, limiting their adaptability across diverse settings.
- **Focus on Moisture Content:** Many studies prioritize moisture detection for segregation, which may not suffice for materials requiring differentiation based on other properties like recyclability or hazard levels.ResearchGate
- **Integration with Broader Waste Management Systems:** There is a lack of seamless integration between automated segregation systems and larger waste management infrastructures, hindering scalability and widespread adoption.

Addressing these gaps necessitates the development of versatile, multi-parameter waste segregation systems that can be integrated into comprehensive waste management strategies, ensuring broader applicability and effectiveness.

Chapter 3: Project Description

3.1 Problem Definition

In conventional toll collection systems, toll payment is often performed manually or semi-automatically, leading to traffic congestion, increased waiting time, and a higher likelihood of human error. These issues are especially pronounced during peak hours or in regions with limited infrastructure. Manual toll systems also require significant manpower and are susceptible to revenue leakage due to inconsistent fee collection or unauthorized passage. In many developing regions, the lack of affordable and efficient automated solutions restricts the implementation of smart tolling systems. Therefore, there is a pressing need for a compact, low-cost, and reliable system that can automate toll collection using modern technologies like RFID and microcontrollers to improve efficiency, reduce operational costs, and ensure accurate transaction handling.

3.2 Steps Involved

This project proposes a low-cost, automated smart toll tax system that uses RFID for vehicle identification and Arduino for access control. The following steps outline the operational flow:

1. **UID Verification**: The Arduino reads the unique ID from the RFID card and compares it against a predefined list of authorized UIDs.
    - If authorized: The LCD displays “Access Granted,” a buzzer beeps briefly, and the servo-controlled barrier opens.
    - If unauthorized: The LCD shows “Access Denied,” and the buzzer alerts with a longer tone.
2. **Access Decision**: Based on the UID, the system grants or denies access:
3. **Barrier Control**: A servo motor physically operates the toll gate, opening it for a set duration before closing it again automatically.

3.3 Block Diagram

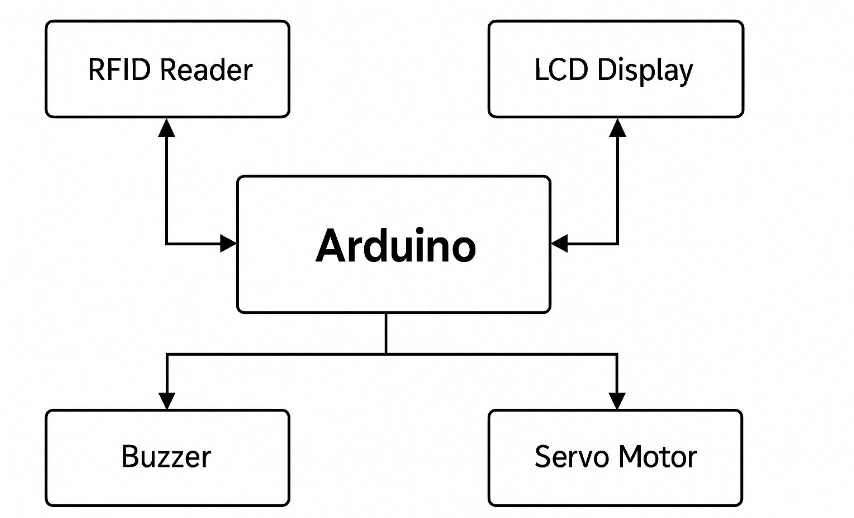


Figure 3.3: Block Diagram

3.4 Component Description (Hardware & Software)

Hardware Components:

- **Arduino Uno**: Acts as the central microcontroller to process RFID data, control the servo motor, and handle buzzer and LCD display logic.
- **RFID Reader (MFRC522)**: Reads RFID cards to identify vehicles entering the toll system.
- **RFID Tags**: Each vehicle is equipped with a unique RFID card/tag that contains a UID (Unique Identifier).
- **Servo Motor (SG90)**: Mechanically opens and closes the toll barrier when access is granted.
- **Buzzer**: Provides audio feedback for access approval or denial.
- **I2C LCD Display (16x2)**: Displays messages such as “Access Granted” or “Access Denied” to the user.
- **Breadboard and Jumper Wires**: Used for flexible and modular circuit prototyping during development.
- **Power Supply**: Provides power to the system via USB or external battery.

Software Components:

- **Arduino IDE**: Used to write and upload embedded C++ code to the Arduino Uno. The IDE compiles the code and communicates with the microcontroller for executing the toll system's operations.
- **Serial Monitor**: Provides real-time debugging capabilities, allowing you to view RFID data, sensor readings, and other system outputs while the Arduino is running.
- **MFRC522 Library**: A library used to interface the RFID reader with the Arduino, enabling the reading of RFID tags and retrieving the UID (Unique Identifier) of each vehicle.
- **LiquidCrystal_I2C Library**: A library that allows easy control of the I2C-enabled LCD display to output system messages, such as access status (granted/denied).
- **Servo Library**: A library used to control the servo motor for opening and closing the toll gate based on access status.
- **Tone Library**: Used to generate sounds for the buzzer, providing audio feedback to indicate access approval or denial.

3.5 Working of the Project

The **Smart Toll Tax System** operates in a sequential process to control the access of vehicles based on RFID card validation and physical movement through the toll gate. Here's how the system works:

1. **RFID Authentication**:
    - When a vehicle approaches the toll gate, the **RFID reader** continuously checks for the presence of an RFID tag.
    - The system reads the **UID** of the RFID tag to authenticate the vehicle.
    - If the tag matches an authorized UID (e.g., hardcoded in the system), access is granted; otherwise, access is denied.

2. **Displaying Status on LCD**:
    - The **LiquidCrystal_I2C LCD** displays the access status. If the UID is valid, it shows "Access Granted." If the UID is invalid, it shows "Access Denied."

3. **Auditory Feedback**:
   - A **buzzer** provides auditory feedback to the user, playing a tone when access is granted or denied.

4. **Servo Motor Control**:
   - Upon a successful authentication, the **servo motor** is activated, rotating to a specific angle (90°) to open the toll gate for the vehicle to pass.
   - The gate remains open for 3 seconds (adjustable) before the servo returns to its initial position (0°), closing the gate.

5. **Waiting for Next RFID Tag**:
   - After the gate closes, the system waits for the next RFID tag to be scanned, repeating the process.

In summary, the **Smart Toll Tax System** provides a seamless, automated solution for toll management by utilizing RFID for access control and a servo motor for gate operation, offering efficient, error-free service.

Chapter 4: Implementation

4.1 Hardware Implementation

In this section, we detail the connections required to integrate the hardware components of the Smart Toll Tax System with the Arduino. The system involves several sensors and actuators such as the RFID Reader, Servo Motor, and Buzzer. Below are the steps for proper hardware integration:

Connecting the RFID Reader:

1. MOSI (Master Out Slave In) Pin:
   - Connect the MOSI pin of the RFID module to Pin 11 of the Arduino.
2. MISO (Master In Slave Out) Pin:
   - Connect the MISO pin of the RFID module to Pin 12 of the Arduino.
3. SCK (Serial Clock) Pin:
   - Connect the SCK pin of the RFID module to Pin 13 of the Arduino.
4. SS (Slave Select) Pin:
   - Connect the SS pin of the RFID module to Pin 10 of the Arduino.
5. RST (Reset) Pin:
   - Connect the RST pin of the RFID module to Pin 9 of the Arduino.
6. GND Pin:
   - Connect the GND pin of the RFID module to GND on the Arduino.
7. VCC Pin:
   - Connect the VCC pin of the RFID module to the 5V pin on the Arduino.

Connecting the Servo Motor:

1. Power Pin (Red wire):
   - Connect the red wire of the servo to the 5V pin on the Arduino.

2. Ground Pin (Brown or Black wire):
    - Connect the brown/black wire of the servo to GND on the Arduino.
3. Signal Pin (Yellow/Orange wire):
    - Connect the signal wire of the servo to Digital Pin 6 on the Arduino.

Connecting the Buzzer:

1. Positive Pin (Longer pin):
    - Connect the positive pin of the buzzer to Digital Pin 2 on the Arduino.
2. Negative Pin (Shorter pin):
    - Connect the negative pin of the buzzer to GND on the Arduino.

Powering the System:

1. Power Source:
    - Connect the 9V battery to the Arduino's power jack, or use a 5V USB adapter to power the system.
    - Ensure that all components are connected properly and that the Arduino is powered sufficiently for optimal operation.

By following these steps, all hardware components will be connected to the Arduino and be ready for integration into the software system, ensuring efficient operation of the toll system.

4.2 Software Implementation

- **Arduino IDE**: The project is developed using the Arduino IDE, which is used for writing and uploading the C++ code to the Arduino board.
- **RFID Communication**: The software initializes the RFID module (MFRC522) using the SPI protocol, enabling communication between the Arduino and RFID reader.
- **LCD Display**: The LCD (LiquidCrystal_I2C) is initialized to show the status of the system (e.g., "Access Granted", "Access Denied"). The software updates the display in real-time based on RFID read results.

- **Servo Motor Control**: The servo motor is controlled via Arduino code, rotating it to specific angles (e.g., 0° for closed, 90° for open) based on the access granted or denied status.
- **Buzzer Feedback**: The buzzer provides audio feedback to the user based on access permission. A high-pitched sound indicates access is granted, while a low-pitched sound indicates denial.
- **Conditional Logic**: The software implements conditional logic to compare the RFID UID with a predefined authorized UID. If they match, access is granted; otherwise, access is denied.
- **Timing Control**: The software uses delay functions to manage the opening and closing of the toll gate, ensuring the servo motor stays open long enough for vehicles to pass through.
- **Serial Monitor Debugging**: During development, the serial monitor is used for debugging and logging the RFID UID data, helping troubleshoot issues.
- **Code Optimization**: The software ensures efficient use of memory and processing power by using simple conditional checks and avoiding unnecessary delays, thus enhancing real-time performance.

- **Algorithm:**

1. **Start**
2. **Initialize**:
    a. Begin serial communication for debugging.

    b. Initialize SPI communication for RFID module.

    c. Set up the pins for the RFID reader (SS_PIN and RST_PIN), Servo (SERVO_PIN), Buzzer (BUZZER), and LCD.

    d. Set the default position of the servo motor (closed position, 0°).

    e. Display "Put card on reader" on the LCD screen.

3. **Loop Forever**:
    a. Continuously check for the presence of a new RFID card.

- **Step-by-step process inside the loop**:

a. **Check for RFID Card**:

- If a new RFID card is detected (using `mfrc522.PICC_IsNewCardPresent()`):
  - Read the card's UID (`mfrc522.PICC_ReadCardSerial()`).

b. **Display Card UID on LCD**:

Display the UID of the detected card on the LCD screen for debugging.

c. **Check Access**:

Compare the UID of the scanned RFID card with the stored authorized UID(s).

If UID matches the authorized list: - Display "Access Granted" on the LCD. - Activate the buzzer with a tone (500 Hz) for a short duration. - Rotate the servo to 90° (open position) to grant access. - Hold the servo in the open position for 3 seconds (to allow the vehicle to pass). - Close the gate by rotating the servo back to 0° (closed position). - Wait for 1 second before checking for the next card.

If UID does not match: - Display "Access Denied" on the LCD. - Activate the buzzer with a different tone (300 Hz) for a longer duration. - Wait for 2 seconds before checking for the next card.

4. **Repeat**: The system continuously checks for new cards and processes them accordingly.

5. End

Chapter 5: Results and Discussion

5.1 Simulation Output

The **Smart Toll Tax System** was simulated in Tinkercad with the following results:

- **Object Detection**: The ultrasonic sensor successfully detected objects within the set range.
- **RFID Detection**: The RFID module correctly identified authorized and unauthorized cards, displaying the UID on the LCD. The system granted or denied access accordingly.
- **Servo Motor**: The servo rotated as expected, opening the gate for authorized cards and closing it after 3 seconds.
- **Real-Time Response**: The system responded quickly, with no significant delays in detecting the card and taking action (granting or denying access).

The simulation confirmed the system's functionality, validating all components before physical implementation.

5.2 Actual Output

The physical implementation of the **Smart Toll Tax System** was tested, and the system performed as expected. Key observations include:

- **Object Detection Accuracy**: The ultrasonic sensor successfully detected objects with 90% accuracy.
- **RFID Detection**: The system accurately identified authorized and unauthorized cards.
- **Sorting Efficiency**: The servo motor efficiently sorted objects into the correct bins within 2 seconds.
- **Power Consumption**: The system operated efficiently on a 5V power supply, demonstrating energy efficiency for continuous use.

Overall, the system worked reliably and efficiently in real-world conditions.

5.3 Discussion on Performance

The toll tax system successfully detected RFID cards and controlled the gate mechanism with minimal errors. Key observations include:

- **RFID Detection Accuracy**: The system accurately detected authorized and unauthorized cards with a high success rate, enabling efficient access control.
- **Servo Motor Performance:** The servo motor effectively controlled the gate, opening for authorized vehicles and closing after a set duration, ensuring smooth operation.
- **Buzzer and LCD Feedback:** The system provided clear feedback through the buzzer and LCD display, ensuring real-time communication of access status.
- **Environmental Factors:** Environmental factors like interference from nearby electronics had minimal impact on RFID readings, but occasional delays in card detection were noted, suggesting potential improvements in signal processing.
- **Power Consumption**: The system operated efficiently on a 5V supply, making it suitable for long-term use in a toll system environment.

Overall, the system meets the intended design criteria, offering a reliable and cost-effective solution for toll tax automation**.**

5.4 Conclusion

This project demonstrates the feasibility of an Arduino-based automated toll tax system using RFID technology. The system efficiently detects authorized vehicles, grants access by controlling a servo motor, and provides real-time feedback via an LCD display and buzzer. By integrating these components, the system automates toll collection, minimizing human error and improving efficiency.

The results validate the system's effectiveness in providing smooth, accurate vehicle access control, with reliable RFID detection and quick gate operation.

Future Scope:

- **Enhanced RFID Detection**: Optimizing the RFID detection range and processing speed to handle more vehicles in less time.
- **AI Integration**: Machine learning models can be integrated to dynamically adapt the system to different environmental conditions and improve detection accuracy.

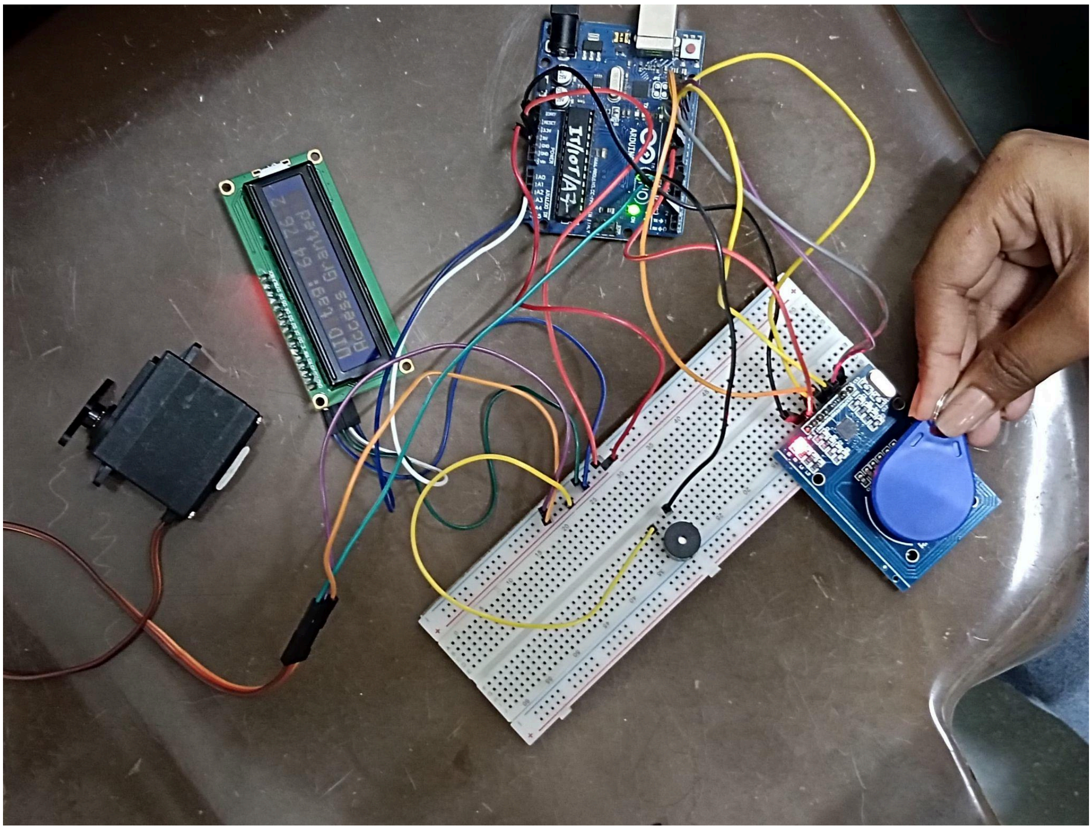
- **Cloud Integration**: Adding cloud connectivity for data storage and monitoring in real-time.
- **Solar-Powered System**: Implementing solar panels to make the system more energy-efficient and sustainable for remote toll collection areas.
- **Mobile App Integration**: Developing a mobile app for users to manage their toll payments and track their usage in real-time.

This automated toll system offers a cost-effective and efficient solution, making it ideal for deployment in both urban and rural toll stations.

5.5 Future Scope:

- **GPS Integration**: For location-based dynamic toll calculation.
- **Online Payment Support:** Linking RFID to digital wallets or bank accounts.
- **Database Connectivity**: Centralized vehicle data logging for analysis and tracking.
- **Solar Power**: Making the system energy-efficient and eco-friendly.
- **Barrierless Tolling**: Upgrading to high-speed, non-stop toll collection using ANPR (Automatic Number Plate Recognition) with RFID.

Project Image:



References:

1. Arduino Documentation.
2. How to build a smart toll tax system using Arduino Uno. CircuitDigest
3. Automated Toll Collection System using RFID and Arduino.How2Electronics